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== Vitis HLS Report for 'matVecMul'
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* Date: Tue May 5 15:50:00 2026

* Version: 2021.1 (Build 3247384 on Thu Jun 10 19:36:07 MDT 2021)

* Project: matVecMul_stream_prj

* Solution: solution_chocolate (Vivado IP Flow Target)

* Product family: zynq

* Target device: xc7z020-clg400-1

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== Performance Estimates
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+ Timing:

* Summary:

Clock	Target	Estimated	Uncertainty
ap_clk	10.00 ns	18.153 ns	2.70 ns

+ Latency:

* Summary:

Latency (cycles)		Latency (absolute)		Interval		Pipeline
min	max	min	max	min	max	Type
1863	3853	33.819 us	69.944 us	1864	3854	no

+ Detail:

* Instance:

Instance	Module	Latency (cycles)		Latency (absolute)		Interval		Pipeline
		min	max	min	max	min	max	Type
grp_matVecMul_Pipeline_sample_loop_fu_251	matVecMul_Pipeline_sample_loop	66	66	1.198 us	1.198 us	66	66	no
grp_matVecMul_Pipeline_matmul_inner_loop_fu_278	matVecMul_Pipeline_matmul_inner_loop	23	24	0.418 us	0.436 us	23	24	no

* Loop:

Loop Name	Latency (cycles)		Iteration	Initiation	Interval	Trip	
	min	max	Latency	achieved	target	Count	Pipelined
- vector_loop	1862	3852	1862 ~ 1926	-	-	1 ~ 2	no
+ matmul_outer_loop	1792	1856	28 ~ 29	-	-	64	no

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== Utilization Estimates
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* Summary:

Name	BRAM_18K	DSP	FF	LUT	URAM
DSP	-	-	-	-	-
Expression	-	-	0	294	-
FIFO	-	-	-	-	-
Instance	4	4	1193	1963	-
Memory	0	-	128	16	0

Multiplexer	-	-	-	237	-
Register	-	-	444	-	-
Total	4	4	1765	2510	0
Available	280	220	106400	53200	0
Utilization (%)	1	1	1	4	0

+ Detail:

* Instance:

Instance	Module	BRAM_18K	DSP	FF	LUT	URAM
control_s_axi_U	control_s_axi	4	0	470	512	0
grp_matVecMul_Pipeline_matmul_inner_loop_fu_278	matVecMul_Pipeline_matmul_inner_loop	0	4	689	1334	0
grp_matVecMul_Pipeline_sample_loop_fu_251	matVecMul_Pipeline_sample_loop	0	0	34	97	0
mux_42_38_1_1_U32	mux_42_38_1_1	0	0	0	20	0
Total		4	4	1193	1963	0

* DSP:

N/A

* Memory:

Memory	Module	BRAM_18K	FF	LUT	URAM	Words	Bits	Banks	W*Bits*Banks
x_localbuff_V_0_U	x_localbuff_V_0	0	32	4	0	16	16	1	256
x_localbuff_V_1_U	x_localbuff_V_0	0	32	4	0	16	16	1	256
x_localbuff_V_2_U	x_localbuff_V_0	0	32	4	0	16	16	1	256
x_localbuff_V_3_U	x_localbuff_V_0	0	32	4	0	16	16	1	256
Total		0	128	16	0	64	64	4	1024

* FIFO:

N/A

* Expression:

Variable Name	Operation	DSP	FF	LUT	Bitwidth P0	Bitwidth P1
add_ln32_fu_327_p2	+	0	0	38	31	1
add_ln44_fu_338_p2	+	0	0	39	32	1
p_Val2_1_fu_429_p2	+	0	0	23	16	16
sub_fu_304_p2	+	0	0	39	32	2
and_ln789_fu_515_p2	and	0	0	2	1	1
and_ln790_fu_529_p2	and	0	0	2	1	1
and_ln795_fu_559_p2	and	0	0	2	1	1
carry_1_fu_449_p2	and	0	0	2	1	1
overflow_fu_553_p2	and	0	0	2	1	1
underflow_fu_577_p2	and	0	0	2	1	1
Rangel_all_ones_fu_481_p2	icmp	0	0	12	14	2
Rangel_all_zeros_fu_487_p2	icmp	0	0	12	14	1
Range2_all_ones_fu_465_p2	icmp	0	0	12	13	2
icmp_ln32_fu_322_p2	icmp	0	0	18	32	32
icmp_ln44_fu_333_p2	icmp	0	0	18	32	32
out_pkt_last_V_fu_357_p2	icmp	0	0	18	32	32
or_ln384_fu_583_p2	or	0	0	2	1	1
or_ln794_fu_541_p2	or	0	0	2	1	1
or_ln795_fu_565_p2	or	0	0	2	1	1
deleted_ones_fu_521_p3	select	0	0	2	1	1
deleted_zeros_fu_493_p3	select	0	0	2	1	1
select_ln384_fu_589_p3	select	0	0	17	1	15

val_V_fu_596_p3	select	0	0	16	1	16
xor_ln416_fu_443_p2	xor	0	0	2	1	2
xor_ln789_fu_509_p2	xor	0	0	2	1	2
xor_ln794_1_fu_547_p2	xor	0	0	2	1	2
xor_ln794_fu_535_p2	xor	0	0	2	1	2
xor_ln795_fu_571_p2	xor	0	0	2	1	2
Total		0	0	294	266	173

* Multiplexer:

Name	LUT	Input Size	Bits	Total Bits
ap_NS_fsm	53	10	1	10
i_fu_172	9	2	31	62
in_stream_TREADY_int_regslice	9	2	1	2
k_reg_240	9	2	32	64
out_stream_TDATA_blk_n	9	2	1	2
x_localbuff_V_0_address0	14	3	4	12
x_localbuff_V_0_ce0	14	3	1	3
x_localbuff_V_0_we0	9	2	1	2
x_localbuff_V_1_address0	14	3	4	12
x_localbuff_V_1_ce0	14	3	1	3
x_localbuff_V_1_we0	9	2	1	2
x_localbuff_V_2_address0	14	3	4	12
x_localbuff_V_2_ce0	14	3	1	3
x_localbuff_V_2_we0	9	2	1	2
x_localbuff_V_3_address0	14	3	4	12
x_localbuff_V_3_ce0	14	3	1	3
x_localbuff_V_3_we0	9	2	1	2
Total	237	50	90	208

* Register:

Name	FF	LUT	Bits	Const Bits
acc_V_0_loc_fu_188	38	0	38	0
add_ln32_reg_662	31	0	31	0
add_ln44_reg_670	32	0	32	0
ap_CS_fsm	9	0	9	0
col_size_read_reg_614	32	0	32	0
grp_matVecMul_Pipeline_matmul_inner_loop_fu_278_ap_start_reg	1	0	1	0
grp_matVecMul_Pipeline_sample_loop_fu_251_ap_start_reg	1	0	1	0
i_fu_172	31	0	31	0
k_reg_240	32	0	32	0
no_of_vectors_read_reg_625	32	0	32	0
or_ln384_reg_700	1	0	1	0
out_pkt_last_V_reg_680	1	0	1	0
overflow_reg_695	1	0	1	0
p_Val2_1_reg_690	16	0	16	0
row_size_read_reg_620	32	0	32	0
select_ln340_1_loc_fu_180	38	0	38	0
select_ln340_2_loc_fu_184	38	0	38	0
select_ln340_loc_fu_176	38	0	38	0
shl_ln_reg_675	6	0	11	5
sub_reg_654	32	0	32	0
targetBlock_reg_685	2	0	2	0
Total	444	0	449	5

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== Interface

* Summary:

RTL Ports	Dir	Bits	Protocol	Source Object	C Type
s_axi_control_AWVALID	in	1	s_axi	control	array
s_axi_control_AWREADY	out	1	s_axi	control	array
s_axi_control_AWADDR	in	14	s_axi	control	array
s_axi_control_WVALID	in	1	s_axi	control	array
s_axi_control_WREADY	out	1	s_axi	control	array
s_axi_control_WDATA	in	32	s_axi	control	array
s_axi_control_WSTRB	in	4	s_axi	control	array
s_axi_control_ARVALID	in	1	s_axi	control	array
s_axi_control_ARREADY	out	1	s_axi	control	array
s_axi_control_ARADDR	in	14	s_axi	control	array
s_axi_control_RVALID	out	1	s_axi	control	array
s_axi_control_RREADY	in	1	s_axi	control	array
s_axi_control_RDATA	out	32	s_axi	control	array
s_axi_control_RRESP	out	2	s_axi	control	array
s_axi_control_BVALID	out	1	s_axi	control	array
s_axi_control_BREADY	in	1	s_axi	control	array
s_axi_control_BRESP	out	2	s_axi	control	array
ap_clk	in	1	ap_ctrl_hs	matVecMul	return value
ap_rst_n	in	1	ap_ctrl_hs	matVecMul	return value
interrupt	out	1	ap_ctrl_hs	matVecMul	return value
in_stream_TDATA	in	32	axis	in_stream_V_data_V	pointer
in_stream_TVALID	in	1	axis	in_stream_V_dest_V	pointer
in_stream_TREADY	out	1	axis	in_stream_V_dest_V	pointer
in_stream_TDEST	in	1	axis	in_stream_V_dest_V	pointer
in_stream_TKEEP	in	4	axis	in_stream_V_keep_V	pointer
in_stream_TSTRB	in	4	axis	in_stream_V_strb_V	pointer
in_stream_TUSER	in	1	axis	in_stream_V_user_V	pointer
in_stream_TLAST	in	1	axis	in_stream_V_last_V	pointer
in_stream_TID	in	1	axis	in_stream_V_id_V	pointer
out_stream_TDATA	out	32	axis	out_stream_V_data_V	pointer
out_stream_TVALID	out	1	axis	out_stream_V_dest_V	pointer
out_stream_TREADY	in	1	axis	out_stream_V_dest_V	pointer
out_stream_TDEST	out	1	axis	out_stream_V_dest_V	pointer
out_stream_TKEEP	out	4	axis	out_stream_V_keep_V	pointer
out_stream_TSTRB	out	4	axis	out_stream_V_strb_V	pointer
out_stream_TUSER	out	1	axis	out_stream_V_user_V	pointer
out_stream_TLAST	out	1	axis	out_stream_V_last_V	pointer
out_stream_TID	out	1	axis	out_stream_V_id_V	pointer